

# Technical Document #637: Computer Vision - Comprehensive Overview

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Object detection identifies and localizes objects in images. YOLO and Faster R-CNN are popular real-time object detection architectures.

Visual question answering (VQA) answers questions about images. It requires both visual understanding and language comprehension.

Image classification assigns a label to an image from a fixed set of categories. CNNs like ResNet and EfficientNet achieve superhuman performance on benchmark datasets.

Image super-resolution reconstructs high-resolution images from low-resolution inputs. GAN-based methods produce photorealistic upscaled images.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

## Key Takeaways:

- Semantic segmentation classifies each pixel in an image into a category.
- Pose estimation detects human body keypoints in images.
- Object detection identifies and localizes objects in images.